The Spherical Pendulum: Cubic Chambers and Focus–Focus Monodromy

HCS Research Program

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Abstract

We give a coordinate-safe analytic atlas for the unit spherical pendulum. Energy–momentum reduction produces an exact cubic $P_{h,j}(u)$, whose discriminant and double-root parameterization delimit the critical-value components. Eight representative regular receipt rows carry independently checked period, azimuthal-angle, and action quadratures. The isolated upright value is focus–focus; with an explicitly fixed oriented basis and matrix-column convention its positive-loop monodromy is $\left[\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right]$. No elementary closed form is asserted for every quadrature, and no arithmetic or Hilbert–Pólya claim is made.

1 Model and a pole-safe reduction

On T^*S^2, in the usual chart away from the poles, let

$$H = \frac{1}{2}\left(p_\theta^2 + \frac{j^2}{\sin^2\theta}\right) + \cos\theta, \qquad J = j = p_\phi. \tag{1}$$

The global embedding variables $r\cdot r = 1,\ r\cdot p = 0$ are retained at $\theta = 0, \pi$, where ϕ is not a coordinate. Put $u = \cos\theta$ and write $h = H$. Hamilton's equations give

$$\dot{u}^2 = P_{h,j}(u) = 2(1-u^2)(h-u) - j^2 = 2u^3 - 2hu^2 - 2u + 2h - j^2. \tag{2}$$

2 Critical-value geometry

The cubic discriminant is

$$\operatorname{disc}_u P = 4(16h^4 - 8h^3j^2 - 32h^2 + 72hj^2 - 27j^4 + 16). \tag{3}$$

For an interior double root $s \in (-1, 0)$, solving $P(s) = P'(s) = 0$ gives

$$h = \frac{3s^2-1}{2s}, \qquad j^2 = \frac{(1-s^2)^2}{-s}. \tag{4}$$

The endpoint $u = -1, h = -1, j = 0$ is elliptic–elliptic. The upright endpoint $u = 1, h = 1, j = 0$ is an isolated focus–focus value and its fiber is pinched; it is not a point on the $s \in (-1, 0)$ interior branch.

Proposition 1 (root chamber)**.** *For each regular receipt row with $j \neq 0$, the three real roots obey $r_1 < r_2 < 1 < r_3$, and the physical oscillation interval is $(r_1, r_2) \subset (-1, 1)$. The endpoint-cancelled substitution $u(t) = m + d\cos t$, $m = (r_1 + r_2)/2$, $d = (r_2 - r_1)/2$, removes the square-root endpoint singularities.*

3 Three exact quadratures

On a regular chamber $P = 2(u - r_1)(r_2 - u)(r_3 - u)$. The period, azimuthal increment and action are

$$T = 2\int_0^\pi \frac{\mathrm{d}t}{\sqrt{2(r_3 - u(t))}}, \qquad \Delta\phi = 2j\int_0^\pi \frac{\mathrm{d}t}{(1-u(t)^2)\sqrt{2(r_3-u(t))}}, \tag{5}$$

$$I = \frac{1}{\pi}\int_0^\pi \frac{d^2\sin^2 t\,\sqrt{2(r_3-u(t))}}{1-u(t)^2}\,\mathrm{d}t. \tag{6}$$

The square-root factor in the numerator of I is essential: it is the Jacobian form of $\pi^{-1}\int_{r_1}^{r_2}\sqrt{P(u)}/(1-u^2)\,\mathrm{d}u$. The producer and an independent SymPy script evaluate both forms at 80–90 digits. The ledger deliberately reports quadratures rather than pretending that all parameters have an elementary closed form.

Table 1: Selected exact-rational controls and certified outputs.

case	(h, j)	r_1, r_2, r_3 (rounded)	T	I
R01	$(-1/2, 1/4)$	$-0.9659, -0.5444, 1.0103$	3.3528	0.1267
R02	$(0, 1/10)$	$-0.9975, -0.0050, 1.0025$	3.7018	0.4879
R04	$(1/4, 1/10)$	$-0.9980, 0.2447, 1.0033$	3.9444	0.6398
R06	$(3/4, 1/20)$	$-0.9996, 0.7472, 1.0025$	4.9260	1.0122

4 Revision-one evidence

The full receipt (eight representative regular rows and seven critical rows) is accompanied by a producer-independent checker with 308 assertions, a 77-check SymPy cross-check, byte replay, and 34 repaired-hash hostile mutations. The original u-integral for I is evaluated independently, so a misplaced square-root factor cannot survive the release gate.

5 Liouville fibers and monodromy

Away from the critical-value set, the commuting pair (H, J) has compact Liouville tori. Fix α as the vanishing cycle and β as a transported complementary cycle. For a positive counterclockwise loop around the isolated value $(h, j) = (1, 0)$,

$$\alpha \longmapsto \alpha, \qquad \beta \longmapsto \beta + \alpha.$$

Our matrix convention is that *columns are transported basis vectors expressed in the initial* (α, β) *basis.* Consequently

$$M_{(\alpha,\beta)} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{7}$$

This convention explains why a row-vector or transposed convention would display a different-looking matrix; no such ambiguity is left in the receipt.

For completeness, a regular torus trajectory closes exactly when $\Delta\phi/(2\pi) = p/q$ in lowest terms. The primitive closure uses q u-oscillations and the k-fold repetition uses kq; irrational ratios are quasiperiodic. The resonant objects remain clean one-parameter torus families, rather than isolated primitive orbits.

References

[1] R. H. Cushman and J. J. Duistermaat, The quantum mechanical spherical pendulum, *Bull. Amer. Math. Soc.* 19 (1988), DOI 10.1090/S0273-0979-1988-15705-9.

[2] H. R. Dullin, Semi-global symplectic invariants of the spherical pendulum, *J. Differential Equations* 254 (2013), DOI 10.1016/j.jde.2013.01.018.